

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Senior Software Engineer, Perception

Senior Software Engineer (C++) with experience shipping real-time vision end-to-end solutions and building asynchronous pipelines with Intel OpenVINO. Delivered safety-critical CV solutions for clients including British Petroleum and Austin Industries under tight latency and power constraints. Background in 3D geometry and SLAM from an MSc Robotics at the University of Bristol, including optimisation of RTAB-Map and LiDAR pipelines for challenging environments. Proven ability to collaborate effectively in cross-functional teams, communicate complex technical concepts clearly, and thrive in fast-paced environments.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Designed and deployed real-time perception pipelines on edge devices using PyTorch and OpenVINO, achieving a 15% F1 score improvement with deterministic 24/7 operation under harsh industrial conditions.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, TensorFlow Model Zoo) and quantify accuracy, latency, and stability pre and post optimisation for regression detection.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Implemented reliability patterns for real-time CV services, including timeouts and retries and Kafka pipelines with message validation to maintain real-time behaviour under load.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Owned end-to-end data pipelines, including automated data collection, augmentation, and labelling using MLflow, DVC, and Optuna, increasing dataset diversity by 40% and reducing manual labelling effort by 60%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.

LivSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Implemented an asynchronous inference pipeline in C++ with OpenVINO, overlapping pre-processing, inference, and post-processing to reduce end-to-end latency and improve pipeline utilisation on Intel GPU targets.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Re-architected the perception pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Integrated Boost C++ libraries into production pipelines, resolving dependency and build issues and maintaining reproducible deployments.

- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map, integrating multimodal LiDAR and radiological data for localisation and mapping in inspection scenarios leveraging the Clearpath Husky development platform.
- Evaluated RTAB-Map (LiDAR-only), LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation prior to real world deployment.
- Implemented non-linear least squares optimisation in C++ using Ceres and Eigen for pose refinement to improve stability under noisy measurements and tuned loss functions and convergence settings to reduce failure cases.
- Validated sensor placements for optimal coverage and calibrated LiDAR extrinsics to the robot base such as the Velodyne VLP 16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation and debugged tf trees and frame connectivity.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

Jan 2020

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulation-only continuation of the dissertation autonomy stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM plus LiDAR+RGB fusion SLAM and navigation.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM in 2D and 3D LiDAR across structured scenarios and implemented autonomous navigation with Nav2.
- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python and implemented tf2 frame trees with message synchronisation and debugged tf connectivity and transform timing issues.
- Implemented point cloud processing using PCL and Open3D for filtering, segmentation and registration to support mapping and localisation stability.
- Implemented an EKF-based localisation layer using robot_localization to fuse wheel odometry and IMU for smooth odom → base_link tracking when running LiDAR-only SLAM, improving motion continuity for Nav2.
- Containerised the stack with Docker on Linux and managed versioning in Git to ensure reproducible builds and consistent execution across development environments.

End-to-End Damage Severity & Claim Assessment

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

SKILLS

- **SLAM:** 3D Feature Extraction, Cartographer, Ceres, Ceres, Convolutional Neural Networks, Data Augmentation, Eigen, GLIM, LiDAR SLAM, LIO-SAM, Object Detection & Tracking, Pose graph optimisation, RTAB-Map, SAM, Scan Matching, Scikit-Learn, U-Net, Vision Transformer (ViT), YOLO
- **Cloud, Edge & MLOps:** Apache Kafka, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Linux, NumPy, Pandas, PostgreSQL, Python, Qt
- **Deep Learning and Optimisation:** Bayesian Optimisation, DeepSORT, Layer fusion, MultiRocket, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Embedded and Low-level:** 8051 Assembly, ADC/DAC interfacing, Arduino C, Embedded C, Interrupt-driven I/O, Timers and counters, UART serial
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics Stack:** Gazebo (Harmonic), Nav2, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), ROS2 image_transport, tf2